

Introduction to Machine Learning

Lab 7: Gaussian Mixture Model for Point Cloud Alignment

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1 Motivation

- Implement the EM algorithm for GMM and get some feelings about the pipeline.
- Based on the EM algorithm, implement a non-trivial application of GMM — affine registration of point clouds.
- Get to know that GMM is not limited to classic clustering problems.

2 Tasks

Please read Lecture 9 carefully before doing this lab work.

- Given a source point cloud $\mathbf{Y} \in \mathbb{R}^{M \times D}$, we would like to align/match it to a target point cloud $\mathbf{X} \in \mathbb{R}^{N \times D}$ via an affine transformation (projection + translation): for each row of $\mathbf{Y}$, i.e., $\mathbf{y} \in \mathbb{R}^D$,

$$\mathbf{x} = \mathbf{A}\mathbf{y} + \mathbf{t} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_D) \quad (1)$$

where $\mathbf{A} \in \mathbb{R}^{D \times D}$ is the projection matrix and $\mathbf{t} \in \mathbb{R}^D$ is the translation vector, σ^2 is the variance of noise.

- Suppose that $\mathbf{X}$ is the observed data of a GMM model with M Gaussian components:

$$\mathcal{N}(\mathbf{A}\mathbf{y}_m + \mathbf{t}, \sigma^2 \mathbf{I}_D), \quad \forall \mathbf{y}_m \in \mathbf{Y} \quad (2)$$

and the components are with the same weight, i.e., $w_m = \frac{1}{M}$ for $m = 1, \dots, M$.

- **Task: Align $\mathbf{Y}$ to $\mathbf{X}$ via an EM algorithm, 1) learn the model parameters $\{\mathbf{A}, \mathbf{t}, \sigma^2\}$ and 2) estimate the correspondence between $\mathbf{Y}$ and $\mathbf{X}$.**
 1. Design the E-step to estimate the correspondence between $\mathbf{Y}$ and $\mathbf{X}$
 2. Design the M-step to learn the model parameters
 3. Design the EM algorithm, including the two modules above and an initialization strategy, to align two point clouds.